



🌐 WORKFLOW ORCHESTRATION

Stop Skipping Airflow. Master DP-700 DAGs.

Configuring Apache Airflow Workspace Settings, Python DAG definitions, Data Factory Pipelines vs Airflow, and the #1 Scheduling Exam Trap.

🌐 Apache Airflow Jobs

⚡ Data Pipelines (GUI)

🎯 Explicit Exam Objective

⚙️ ORCHESTRATION TOOLS AT A GLANCE

Exam Domain 1

DAG

Apache Airflow Jobs

Python code-first workflow orchestration in Fabric

GUI

Data Factory Pipelines

Visual drag-and-drop activities & event triggers

100%

DP-700 Exam Aligned

Covers updated July 2026 exam study guide



TOOLING OVERVIEW

4 Orchestration Tools in Microsoft Fabric

Understanding when to use visual pipelines, Airflow DAGs, notebooks, or Dataflows.



1. Data Pipelines

Visual drag-and-drop ETL orchestration. Supports schedules, storage events & Webhooks.



2. PySpark Notebooks

Code transformation engine. **CANNOT self-schedule directly** (Needs Pipeline or Airflow).



3. Dataflows Gen2

Visual Power Query M transformation engine. Managed refresh schedule.



4. Apache Airflow

Python code-first DAG orchestration. Managed workspace level execution.



DP-700 Core Exam Objective

"Configure Apache Airflow workspace settings" is explicitly listed under **Domain 1: Implement & Manage Analytics (30–35%)**. Workspace admins configure Airflow pools, environments & requirements centrally.



CONFIGURATION BOUNDARY

Where to Configure **Apache Airflow**?

Apache Airflow in Fabric is managed at the **Workspace Settings** level.

NAVIGATION PATH IN FABRIC UI



LOCATION

Workspace Settings → Data Engineering → Apache Airflow



AIRFLOW REQUIREMENTS

Upload custom `requirements.txt` for PyPI DAG dependencies



CONNECTIONS & SECRETS

Store Azure Key Vault & Airflow Connections securely

WHAT AIRFLOW SETTINGS ARE NOT



NOT TENANT-WIDE

Airflow configuration does NOT apply globally across the entire tenant



NOT ITEM-LEVEL ONLY

Environment settings and requirements are set at Workspace boundary



EXAM TRAP #1

The #1 Notebook Scheduling Trap

A critical distinction between standalone notebooks and pipeline orchestration.



The Common Trap

"You can configure a native cron schedule directly inside a PySpark Notebook item settings."



The Correct DP-700 Answer

PySpark Notebooks **cannot self-schedule directly**. You MUST wrap the notebook in a **Data Factory Pipeline** or an **Airflow DAG** to trigger automated runs!



Why This Matters on the Exam

Multiple-choice questions will offer "Schedule the notebook item directly" as a distractor option. **Always eliminate this option immediately!** Choose Data Pipelines or Airflow.

[Data Engineering](#) > [DP-700](#) > [DAG Definition](#)

CODE-FIRST WORKFLOW

Fabric Airflow DAG Definition

Creating Python-based Directed Acyclic Graphs (DAGs) to orchestrate Fabric pipelines and notebooks.

 EXAMPLE: AIRFLOW DAG ORCHESTRATING FABRIC NOTEBOOK ACTIVITY

```
from airflow import DAG from airflow.operators.python import PythonOperator from datetime import datetime, timedelta default_args = { 'owner': 'fabric_data_engineer', 'start_date': datetime(2026, 7, 21), 'retries': 2 } with
DAG('fabric_daily_etl_dag', default_args=default_args, schedule_interval='@daily') as dag: task_ingest = PythonOperator(task_id='run_pipeline', python_callable=trigger_fabric_pipeline) task_transform =
PythonOperator(task_id='run_notebook', python_callable=trigger_fabric_notebook) task_ingest >> task_transform # Set dependency pipeline → notebook
```

ARCHITECTURE COMPARISON

Data Pipelines vs Apache Airflow

Comparing low-code visual pipelines with code-first Python DAG orchestration.

⚡ DATA FACTORY PIPELINES

-  INTERFACE
Visual Drag-and-Drop GUI
-  TRIGGERS
Schedule, Storage Event, Webhooks
-  BEST FOR
Standard ETL, high-throughput Copy Activity

⚙️ APACHE AIRFLOW JOBS

-  INTERFACE
Python Code (DAG Files)
-  DEPENDENCIES
Complex cross-system DAG branching & backfills
-  BEST FOR
Multi-cloud workflows & existing Airflow migrations



PIPELINE EXECUTION

Pipeline Triggers & Activities

Orchestrating notebooks, Dataflows, and webhooks with flexible execution triggers.



Schedule Trigger

Run pipeline on fixed time intervals (hourly, daily, weekly) or cron expressions.



Storage Event Trigger

Trigger pipeline automatically when a new file lands in OneLake or ADLS Gen2 container.



Webhook / REST API

Invoke Fabric pipelines remotely via Azure Logic Apps, Azure Functions, or external APIs.



EXAM DECISION GUIDE

Orchestration Tool Cheat Sheet

Quick reference guide to select the right orchestration tool for DP-700 scenarios.

SCENARIO SELECTION CHEAT SHEET

SCENARIO REQUIREMENT

Migrating existing Python DAG workflows from Astronomer / MWAA

Apache Airflow Job

SCENARIO REQUIREMENT

Schedule a PySpark notebook to execute daily at 2:00 AM

Data Factory Pipeline

SCENARIO REQUIREMENT

Trigger ETL pipeline immediately when a CSV file arrives in OneLake

Storage Event Trigger

SCENARIO REQUIREMENT

Configure PyPI libraries for Airflow DAGs across a workspace

Workspace Settings



TIP #4 OF 30

Remember: Airflow is **Workspace Level**

In DP-700 questions, pay close attention to configuration boundaries. **Apache Airflow pools, environments, and requirements.txt are configured at Workspace Settings** — NOT at the Tenant Admin level!

Workspace Boundary

Python requirements.txt

Exam Favorite



DAY 4 COMPLETE

Day 4 — Done!

You now master **Apache Airflow Jobs**, Workspace Settings, Pipeline Triggers, and Notebook Orchestration rules for DP-700.

Follow on LinkedIn for Day 5 →

Tomorrow: Lakehouse — Files vs Managed Delta Tables & SQL Analytics Endpoint

ABOUT THE AUTHOR



Sayan Banerjee

Databricks Certified Data Engineer • Microsoft Fabric Data Engineer (DP-700) • Power BI Analyst (PL-300)
AI, BI & Analytics Consultant • Embee Software • Kolkata